

VICTOR HUGO CAMPOS

Forward Deployed Engineer · Multi-Agent AI in Production · Healthcare & Enterprise

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SUMMARY

Engineer who builds AI agents and then owns them in production. I currently work as the sole engineer on a multi-agent conversational platform serving 7,000+ students, where I do the full forward-deployed loop myself: scoping loosely defined ideas with non-technical stakeholders, translating them into agents, instrumenting them to prove they behave, and debugging live production issues raised by the users themselves. Before that I spent a year in healthtech on a B2B platform with 50K+ active users, working inside the constraints that come with health data and enterprise clients. Comfortable holding a conversation with a non-technical champion in the morning and debugging a production pipeline in the afternoon.

EXPERIENCE

Software & AI Engineer | Forward Deployed Engineer · Barkus (EdTech)

May 2026 – Present

Remote · Contract · Sole engineer on lara 2.0

- **Ownership:** Sole engineer behind a WhatsApp-native multi-agent AI platform in production with 7,000+ public and private school students and 1.5M+ messages processed. Owned architecture, implementation, infrastructure and production support end to end.
- **Forward deployed:** I am the person the operations team and the students talk to when something breaks. I diagnose and fix issues directly in production, then feed what I learn back into the platform — the loop from "a user says this is broken" to "the platform no longer works that way" closes in days.
- **Stakeholder scoping:** Work directly with the pedagogical and operations teams to turn loosely defined ideas into concrete agent behavior, defining what success actually measures before building.
- **Multi-agent system:** Architected a pipeline of specialized agents coordinated by a state machine that drives the full learning journey. Each agent has dedicated system prompts, context injection and tool access; messages route to the right agent based on state, progress and conversation history.
- **Behavioral personalization:** Built a system that dynamically adapts tone, difficulty, content framing and gamification for each student in real time.
- **Observability:** Instrumented every LLM call with Langfuse to trace agent behavior per message, catch regressions and drive continuous cost optimization in production (prompt caching, sampling, elimination of wasteful retries).
- **Zero-downtime migration:** Led the removal of Chatwoot from the architecture, rebuilding the messaging pipeline directly against Meta's WhatsApp Cloud API with zero downtime and zero lost messages, fronted by an async Cloud Tasks queue with retries and a DLQ.
- **Core engineering:** Three independent backend services — AI engine (Express + raw SQL), CMS with an AI content-authoring pipeline (Next.js 14 + Prisma), and teacher portal (Fastify 5 + JWT, rankings, audit trail, CSV/Excel export) — over a multi-tenant PostgreSQL schema with full tenant isolation.
- **Support tooling:** Built lara's Support Center, used daily by the non-technical support team to resolve operational issues without engineering involvement.

Stack: TypeScript · Express · Next.js 14 · Fastify 5 · Prisma · PostgreSQL 16 · Google ADK · OpenAI GPT-4.1 · Gemini 2.5 Flash · Whisper · GCP Cloud Run, Cloud SQL, Cloud Tasks, Cloud Build · Langfuse · Meta WhatsApp Cloud API

Software & AI Engineer · Independent / Contract

Mar 2026 – Present

São Paulo, Brazil & Remote · AI product delivery for startups and enterprise clients

- **Multi-agent video generation SaaS:** Platform that generates complete commercial videos from a text briefing using 12 specialized agents orchestrated in LangGraph, connected in a graph with persistent state and failure recovery. Built quality gates where critic agents score each phase and trigger automatic regeneration, plus a self-healing mechanism that detects safety blocks, timeouts and failures and regenerates artifacts without human intervention.
- **AI B2B prospecting platform:** Full-stack prospecting and lead-qualification product with a two-stage domain enrichment pipeline (Apify actor + AI verifier agent), an ICP system with AI-generated targeting, and LinkedIn prospecting on an async queue.
- **AI-powered LMS:** Designed and delivered a complete learning platform for a PhD professor at a leading Brazilian university — sole engineer across frontend, backend and infrastructure. GPT-4o quiz generation from PDFs with validatable structured output, a learning-gap detection pipeline returning personalized diagnostics, and multi-role access control with Argon2id + server-side pepper.
- **Conversational agents in hospitality:** Built and deployed a WhatsApp lead-qualification agent and a RAG-based internal concierge for a luxury hospitality brand, letting non-technical staff answer guest questions instantly.
- **Agent-driven development:** Since this engagement I work spec-driven and agent-oriented: spec first, then harness engineering — context design, tool access and verification loops — with implementation delegated to coding agents. Daily use of Claude Code, Claude Cowork and Antigravity.

Stack: TypeScript · Python · LangGraph · Agno · Next.js · NestJS · Fastify · BullMQ · Redis · Postgres · pgvector · Supabase · Docker Swarm · Traefik · ffmpeg · Imagen · Veo · Claude · Gemini · OpenAI

Full-Stack Developer (Trainee) • SysMap Solutions — Fully Ecosystem (Healthtech)

Apr 2025 – Mar 2026

São Paulo, Brazil (Remote) • Client: Fully Ecosystem — Prudential B2B health platform

- **Healthcare at scale:** Production platform with 50K+ active and 100K+ total users across 3 languages, backed by a BRL 200M investment from Prudential, serving B2B health and wellness solutions.
- **Legacy internalization:** Contributed to bringing critical outsourced modules in-house, including the user digital wallet and financial transaction processing — part of an initiative that generated millions of BRL in operational savings.
- **Digital wallet:** Built the core backend of the user financial module with NestJS and TypeORM: monetary transactions, balance management and payment flows.
- **Outcome ownership:** Raised the iOS store rating from 2.8 to 4.3 and Android from 2.5 to 4.9 through continuous technical improvement, performance work and UX-focused engineering.
- **Engineering standards:** Created and documented the unit and integration testing framework with Jest, adopted company-wide as the official internal guide. Built AWS Lambda automations for operational and analytical reporting.

Stack: NestJS • Node.js • Kotlin • TypeORM • Jest • Docker • AWS (Lambda, S3, ECS)

Tech Lead & Full-Stack Developer • Freelance

Jun 2024 – Apr 2025

Remote

- Led end-to-end delivery of digital products in innovation and luxury hospitality, owning architecture and technical decisions — a multilingual conversion-focused website (PT/EN/ES) and full-stack feature development on a hackathon and open innovation platform.

Front-End Developer • Elastic Code

Jan 2024 – Jun 2024

Remote • Internship

- Led the frontend architecture of a Fleet and Freight Management ERP — driver management, cost/weight reporting and real-time dashboards — managing global state across unrelated views with Redux.

Earlier

Apr 2023 – Jan 2024

- Freelance Shopify e-commerce development (conversion-focused storefronts in Liquid) and a first frontend internship at Campos Developers building CRUD interfaces and REST integrations with Vue 2 / Nuxt.

EDUCATION

Associate Degree in Systems Analysis and Development • FIAP

2024 – 2025

- Project-based learning with real projects for partner companies. Led an ERP for iFood focused on financial management and data visualization, and a privacy-preserving employee sentiment platform with Softtek.

AWARDS & LEADERSHIP

- **Regional Winner & Global Nominee — NASA Space Apps Challenge (2023):** Led development of a social network for scientists in 48h, in a hackathon run with the Brazilian Space Agency (AEB). Recognized globally among thousands of teams.
- **2nd Place Nationally — CAPES Hackathon (2024):** Led Brazil's 2nd best solution for the CAPES Academic Portal — AI-integrated search and intelligent assistant. 53 teams, 519 participants.
- **Technology & Innovation Leader — Aspire Leaders Program:** Program founded by Harvard Business School professors, present in 195+ countries.

TECHNICAL SKILLS

AI Engineering: Multi-agent systems, agent orchestration, state machines, LangGraph, LangChain, RAG, prompt engineering, LLM observability (Langfuse), AI cost optimization, OpenAI API, Gemini API, Whisper ASR, FAISS, pgvector, Agno

Agent-Driven Development: Spec-driven development, harness engineering, agent-oriented workflows — Claude Code, Claude Cowork, Antigravity (daily)

Backend: Node.js, TypeScript, NestJS, Express, Fastify, Python, Kotlin, Java, Prisma, TypeORM, raw SQL

Frontend: React, Next.js, React Native, Tailwind CSS, Redux, Styled Components

Cloud & Infra: GCP (Cloud Run, Cloud SQL, Cloud Tasks, Cloud Build), AWS (Lambda, S3, ECS), Docker, Docker Swarm, Traefik, Kubernetes, CI/CD, Cloudflare

Data: PostgreSQL, MongoDB, multi-tenant schema design, data modeling, PL/SQL

Practices: Production support & incident debugging, testing (Jest, TDD, integration), DDD, Clean Architecture, code review, Scrum

Certifications: AWS Certified Developer – Associate • AWS Certified Cloud Practitioner • AWS re/Start • LangChain: Develop AI Agents

Languages: Portuguese (native) • English (professional working proficiency)